

Characterisation and use of energy flexibility in water pumping and storage systems

Rui Amaral Lopes^{a,b,*}, Rune Grønberg Junker^c, João Martins^{a,b}, João Murta-Pina^{a,b}, Glenn Reynders^{d,e}, Henrik Madsen^c

^a NOVA School of Science and Technology, NOVA University of Lisbon, Portugal

^b Centre of Technology and Systems/UNINOVA, Portugal

^c Department of Applied Mathematics and Computer Science, Technical University of Denmark (DTU), Denmark

^d VITO, Unit Smart Energy and Built Environment, Belgium

^e EnergyVille, Cities in Transition, Belgium

HIGHLIGHTS

- Energy flexibility in water pumping and storage systems is explored.
- Proposed methodology can be used to support the operation of power systems.
- A real water distribution system is considered to assess the proposed methodology.
- Energy costs and pumping cycles are reduced by 16% and 57%, respectively.

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ABSTRACT

Renewable energy integration in power systems and increasing electrification of energy demand create new challenges to which energy flexibility can provide effective solutions. Through an innovative use of cumulative energy consumption curves, which represent the maximum and minimum energy limits, as well as the associated flexible energy consumption, this paper presents a methodology to characterise and use the energy flexibility provided by water pumping and storage systems (WPSS) in order to achieve specific objectives at different levels of power systems. The methodology is applied to a case study considering a real WPSS where energy flexibility is used to reduce electricity costs and support the operation of the power system during a wind generation curtailment event. Collected results show that savings around 16% can be achieved while reducing pumping cycles by 57%. Furthermore, the WPSS operation can be modified according to the needs of the power system using the available energy flexibility.

1. Introduction

Renewable energy integration in the European Union (EU) energy system is a key factor to comply with the 2015 Paris agreement on climate change [1] and the EU policy framework for climate and energy [2]. As a result, ambitious targets are being set, not only at the European Commission (EC) level but also by some EU countries. For instance, recent plans defined for Portugal establish that 47% of the final energy consumption should come from renewable energy by 2030 [3]. To achieve this, renewable energy sources should cover 80% of electrical energy production, 38% of heating and cooling energy needs and 20% of

the energy required for transportation. Similar approaches are being followed by other countries [4]. To accommodate the time variability of renewable energy resources and the increased electrification of energy demand, existing power systems should evolve and be prepared to mitigate possible negative impacts. Energy system analysis for the scenarios based on the EU Energy Roadmap 2050 shows that future needs for grid investment, mainly for the distribution grid reinforcement, are essentially driven by the substantial growth and the large-scale deployment of distributed (renewable) energy generation [5]. It also shows that demand response is one of the most promising low-cost instruments to drastically reduce the cost of integrating distributed

* Corresponding author at: NOVA School of Science and Technology, NOVA University of Lisbon, Portugal.

E-mail address: rm.lopez@fct.unl.pt (R. Amaral Lopes).

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renewable energy sources. Moreover, European Energy Regulators consider that a high-level, harmonised, principles-based approach to underpin the regulatory framework concerning facilitating energy flexibility is the optimal way to deliver a more affordable, secure and efficient electric system [6].

In this context, besides resources optimization at the supply-side [7,8], energy flexibility available at the demand-side should be characterised and used to support power systems operation while respecting the needs of the entities providing such flexibility. This strategy is in line with the definition introduced by Annex 67 of the International Energy Agency's Energy in Buildings and Communities Programme, which states that "*the energy flexibility of a building is the ability to manage its demand and generation according to local climate conditions, user needs, and energy network requirements*" [9]. At the building-level, several methodologies to characterise energy flexibility can already be found in the literature [10–17]. A detailed review of these and other methodologies has been conducted in [18]. That review concluded that while these methodologies differ in several aspects, three common properties are identified, namely: i) the temporal energy flexibility; ii) the rate at which energy consumption changes are made; and iii) the costs of using energy flexibility. Temporal flexibility is defined as the ability of a system to shift its power demand or supply in time. The rate of energy consumption changes quantifies the potential to modify the amplitude of the power demand during a certain period. The cost of using energy flexibility quantifies the deviation of the system from its normal operation to unlock the energy flexibility towards the grid. This deviation often implies an economic cost (e.g., due to storage losses).

Nevertheless, to further exploit energy flexibility available at the demand-side, through local (e.g., community) or large-scale (e.g., country) aggregated use, the contribution of other non-building flexible systems should also be considered. This paper presents the case of drinking water pumping and storage systems (WPSS) integrating municipal water distribution networks, whose operation, and respective energy flexibility, can be considered together with other systems (e.g., buildings) to achieve specific objectives such as renewable energy integration or peak load reduction. At the same time, WPSS operators are interested in the internal use of available energy flexibility to reduce their operational costs since the energy required to collect, treat and pump water represents 30–40% of municipal energy bills in the EU [19]. Taking this into consideration, there is a need to characterise and use the

energy flexibility provided by WPSS both internally (e.g., to reduce costs) and externally (e.g., to support grid operation), which is addressed in this paper.

A typical drinking water cycle (Fig. 1) begins with the water collection from different sources to be processed in water treatment facilities. Downstream pumping stations are used to distribute the drinking water to smaller reservoirs closer to final consumption centres where the water is then distributed to final consumers [20]. While strict water security rules must be respected [21], including minimum and maximum allowed water tank levels, the operation of these pumping stations can be controlled to take advantage of the final reservoirs storage capabilities in order to, e.g., reduce electrical energy costs [22]. As such, energy flexibility can be provided by combining the pumping station operation and the storage capacity of the final reservoirs, which form a WPSS as illustrated in Fig. 1.

A methodology to characterise and tackle the internal/external use of the energy flexibility provided by WPSS is described in Section 2. To illustrate the application of this methodology, a real WPSS, located in the centre of Portugal, is used as a case study in Section 3. The available energy flexibility is used to i) reduce electrical energy costs given the existing time-of-use tariff (internal use); and ii) support the grid operation during a time period with excessive wind energy availability (March 12 and 13, 2017), which resulted in the first wind generation curtailment event registered in Portugal (external use). Conclusions and future work are addressed in Section 4.

2. Methodology

In this work, the predicted cumulative energy consumption of WPSS is used to characterise the available flexibility over a time period T , as conceptually represented in Fig. 2. This method can only be applied under the strict conditions of linear systems, for which storage losses are negligible [6], and it will be shown that WPSS comply with these strict requirements.

In Fig. 2, $E_{MAX}(n)$ and $E_{MIN}(n)$ represent the maximum and minimum possible cumulative energy consumption profiles between time-steps 0 and T , which are obtained using the forecasted water demand profile and a model of the storage reservoir that provides the water level, $L(n)$, for a specific time-step n , according to the inlet (W_{in}) and the outlet (W_{out}) water volumes relative to time-step $n-1$ and the water level at

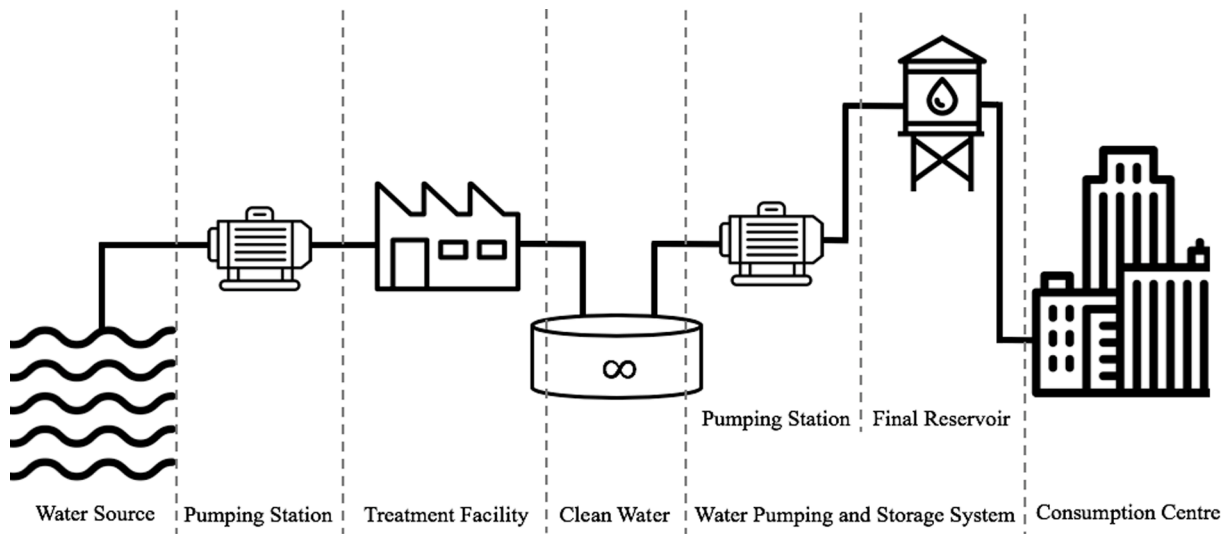


Fig. 1. Illustrative representation of a typical drinking water supply system.

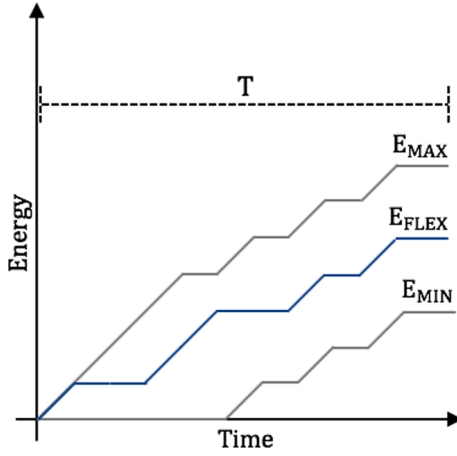


Fig. 2. Energy flexibility representation using cumulative energy consumption curves.

$n - 1$. In this study $L(n)$ is given by:

$$L(n) = L(n - 1) + a_0 + a_1 \times (W_{in} - W_{out}) \quad (1)$$

where coefficients a_0 and a_1 can be found by minimizing the difference between the model output and the real water level, as described in Section 3.

The maximum energy consumption profile, $E_{MAX}(n)$, is obtained by forcing the operation of the pumping station whenever the water level is below the upper limit, hence sustaining the maximum water level in the reservoir. Regarding $E_{MIN}(n)$, the minimum energy consumption results from forcing the pumping station to stop whenever the water level is higher than the lower limit. Both $E_{MAX}(n)$ and $E_{MIN}(n)$ are therefore obtained by integrating the power demand of the WPSS during the period of analysis. In this study, the WPSS presents a binary power demand (zero or maximum value), as described in Section 3. In general terms, the WPSS operation can be controlled in order to achieve any cumulative energy consumption profile $E_{FLEX}(n)$ that falls within the constraints of $E_{MAX}(n)$ and $E_{MIN}(n)$ and respects the possible power values of the pumping station $P_{MIN}, \dots, P_{MAX}$.

To make optimal use of the available energy flexibility, the Water System Operator (WSO) should find the profile $E_{FLEX}(n)$ that solves the problem described by Expressions (2)–(5), where $Penalty(n)$ refers to the penalty signal imposed to the WPSS operation for each unit of consumed energy. As described in [23], penalty signals can be designed in order to achieve different objectives, like adapting the energy consumption to wind or solar generation or to support grid operation during periods of excessive load. This problem should be solved at each time-step, n , with the result being the power required by the pumping station between two consecutive time-steps, which is given by $[E_{FLEX}(n) - E_{FLEX}(n - 1)] / \Delta t$, where Δt refers to the time period between the two consecutive instants.

$$\min_{E_{FLEX}} \sum_{n=1}^T [E_{FLEX}(n) - E_{FLEX}(n - 1)] \cdot Penalty(n) \quad (2)$$

s.t.

$$E_{FLEX}(n) \leq E_{MAX}(n), \forall n \in T \quad (3)$$

$$E_{FLEX}(n) \geq E_{MIN}(n), \forall n \in T \quad (4)$$

$$\frac{[E_{FLEX}(n) - E_{FLEX}(n - 1)]}{\Delta t} \in P_{MIN}, \dots, P_{MAX}, \forall n \in [1, T] \quad (5)$$

The most common example of a penalty signal is the electricity price. Using dynamic tariffs the electricity price can assume, for instance, different values throughout the day and season (winter/summer). The

available energy flexibility is then used internally to reduce the electricity bill over a specific period (i.e., the cumulative penalty). Nevertheless, an external entity can also require the use of the available energy flexibility to improve its operation. In this case, the external entity requires information on $E_{MAX}(n)$, $E_{MIN}(n)$ and $P_{MIN}, \dots, P_{MAX}$, which are provided by the WPSS, in order to find the most suitable profile $E_{FLEX}(n)$. The fact that the external entity does not require any specific knowledge of the WPSS but only the referred profiles and power values represents the strongest feature of this approach.

3. Baseline operation of the water pumping and storage system

A WPSS belonging to a municipal water distribution system in the centre of Portugal is considered in this study. This WPSS comprises a 150-kVA pumping station and a reservoir with a capacity of 6000 m³ (maximum water level of 4.9 m). Water is pumped from a lower elevation reservoir considered to have infinite capacity. During normal operation, the pumping station relies on a single 75-kVA pump responsible for a water inlet of 86 m³ per 15 min of continuous operation. A second 75-kVA pump is available for redundancy and providing backup when the demand is too high for a single pump to keep water level above safety levels. In terms of active power, both pumps show a consumption of 68 kW. Pumps are also cycled to reduce start/stop operations and minimize wear. The hydraulic diagram is shown in Fig. 3.

A Supervisory Control And Data Acquisition (SCADA) system allows monitoring all relevant variables, namely the water level, the water inlet and outlet flow rates, and the electric power supplied to the pumps. The SCADA system also allows pump control through a rule-based approach defined by the WSO. These rules allow managing the operation of the pumps, aiming at maximizing energy consumption during the cheapest periods. The WSO has a weekly-cycle contract (time-of-use tariff periods are defined according to business days, Saturday and Sunday), with four distinct tariff periods, which are, from the most expensive to the cheapest one, peak, half-peak, normal off-peak and super off-peak. There is a minimum level at which the operational pump is started, and a maximum level where it is stopped for each of these tariff periods, that were defined based on the empirical experience of the local operators. There is also a minimum-security level at which, independently of the tariff period, the second pump is started to compensate high demand while maintaining a safety stored volume. The ranges of these levels are shown in Fig. 4, together with the minimum and maximum allowed levels (dashed lines). Off-peak, shown in Fig. 4 comprises the levels associated with the rules applied during both normal and super off-peak periods.

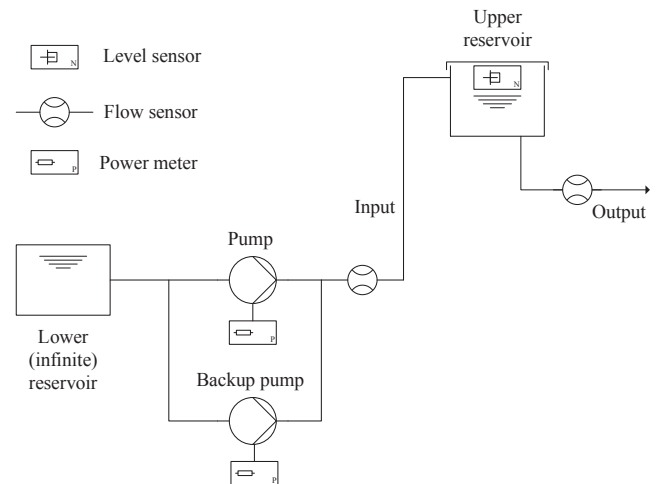


Fig. 3. Hydraulic diagram of the studied WPSS.

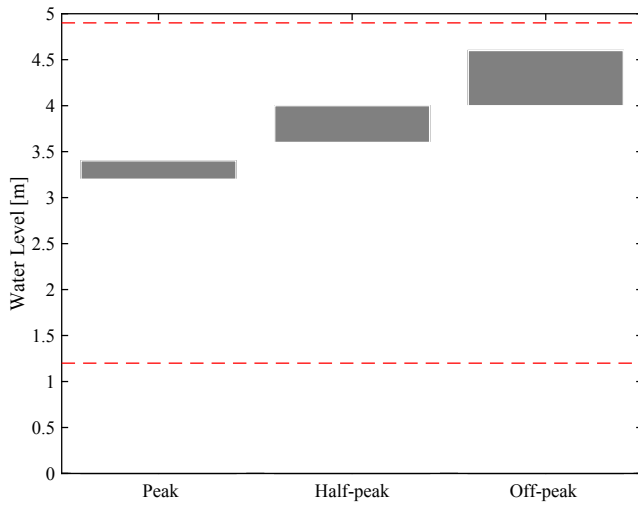


Fig. 4. Ranges of levels for pump operation according to time-of-use tariff periods. Dashed lines denote maximum and minimum level limits.

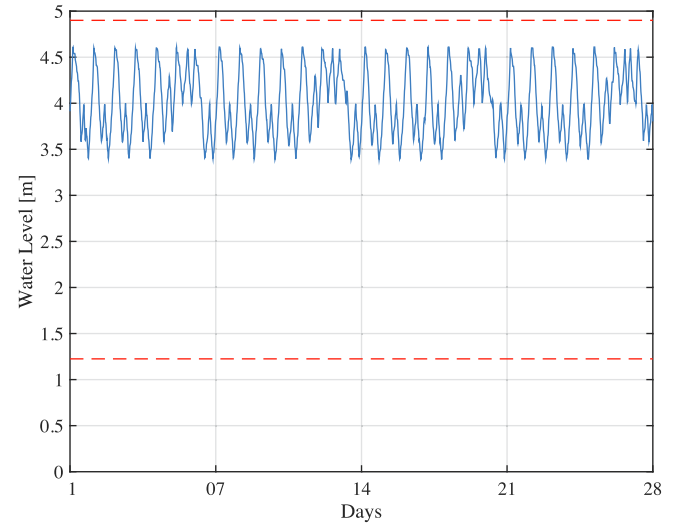
The effects of the pump operation strategy based on the above-mentioned rules can be seen in Fig. 5.a), representing the baseline scenario for this work (from February 28 to March 27, 2017). Sundays, where there are no peak and half-peak periods, are easily identifiable, as according to the rules illustrated in Fig. 4 the water level is maintained above 4 m. Although the demand is met, this strategy leads to an extremely intermittent pump operation, which is shown in Fig. 5.b), and whose excessive number of starts/stops events are adverse to its lifetime.

The water demand profile for the baseline period, also obtained from the referred SCADA system, is shown in Fig. 6.a) considering a 15-min time resolution. The average daily profile of that period is shown in Fig. 6.b), where the typical minimum demand in the night period, as well as the maximum demands around noon and dinner times, are also distinguishable.

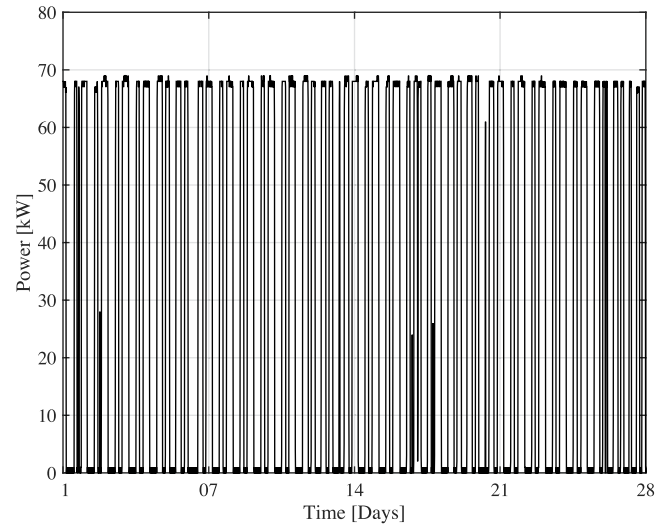
The coefficients a_0 (4.40×10^{-4} m) and a_1 (9.62×10^{-4} m⁻²) of the water level model (see Section 2) were obtained by minimizing the difference between the model output and the real water level presented in Fig. 5.a), when considering the water input associated with the operation of the pumping station (Fig. 5.b)) and the water demand (Fig. 6). The error between the model output and the real data is relatively low (less than 1% for each time-step of the period of analysis), as it can be observed in Fig. 7.

4. Using the available energy flexibility

The rule-based control of the WPSS described in Section 3 uses part of the available flexibility to reduce electrical energy consumption during the most expensive periods. These rules were designed according to the experience of the operators and, as described by them, can result in significant savings when compared to a strategy where only a single maximum/minimum water level limit is considered. Nevertheless, results presented and analysed in Subsection 4.1 show that the internal use of energy flexibility can lead to further savings. Subsection 4.2 illustrates how the available energy flexibility can be used by an external entity to support grid operation. All results consider the water demand profile



a)



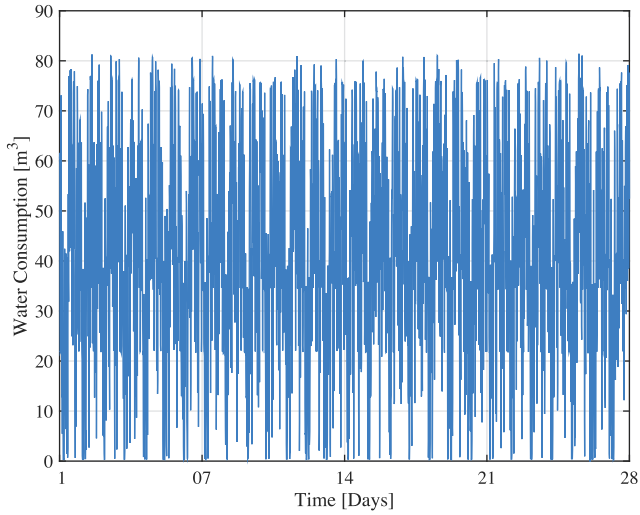
b)

Fig. 5. Baseline operation from 28/02/2017 to 27/03/2017. a) tank water level. b) power consumption.

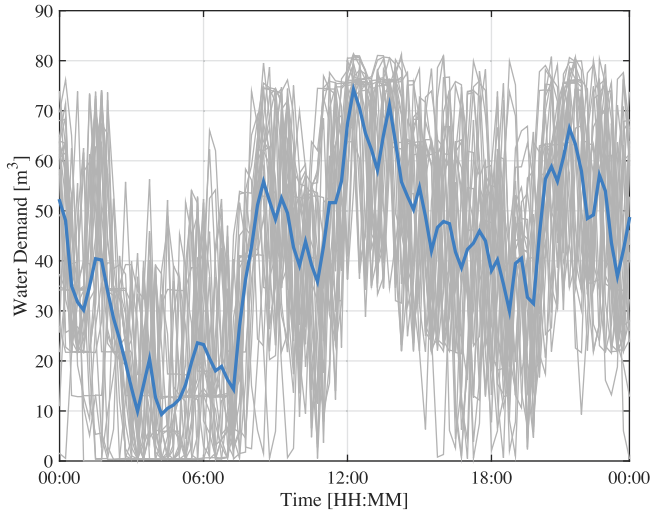
and WPSS characteristics presented in Section 3.

4.1. Internal use of energy flexibility to reduce electricity costs

The benefits resulting from the internal use of energy flexibility from 28/02/2017 to 27/03/2017 are quantified by comparing the obtained results with the costs reported in the electricity bill provided by the WSO for the same period. As presented in Fig. 8, the electricity price, which is used as penalty signal in this internal use of energy flexibility, varies throughout the week with specific tariff periods for Business Days (BD), Saturdays (SAT) and Sundays (SUN). Green, yellow and red background colours in Fig. 8, and in others throughout this section, are associated with off-peak, half-peak and peak periods, respectively. The presented values were extracted from the electricity bill provided by the WSO and



a)



b)

Fig. 6. Water demand from 28/02/2017 to 27/03/2017. a) 15-min resolution raw data. b) daily profiles (grey) and daily average profile (blue).

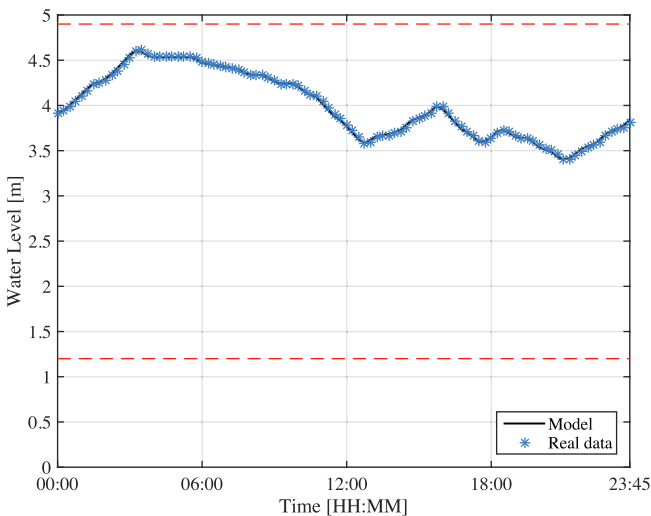


Fig. 7. Output of the water level model and respective real data for 28/02/2017.

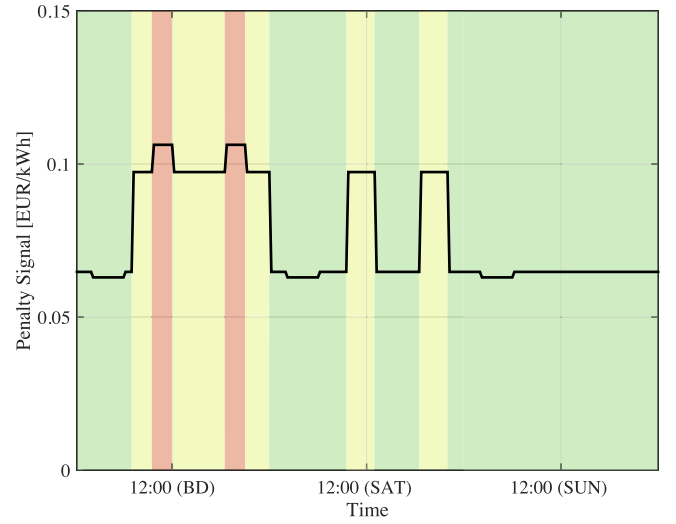


Fig. 8. Penalty signal for the internal use of energy flexibility.

consider the cost of the consumed energy and the cost of using the distribution grid (before taxes), both in €/kWh. Appendix A details the time-of-use periods and tariffs used to produce Fig. 8. While the tariff periods are set according to the Portuguese legislation [24], the respective electricity costs result from negotiation between the WSO and the electricity providers, which in Portugal operate in open market.

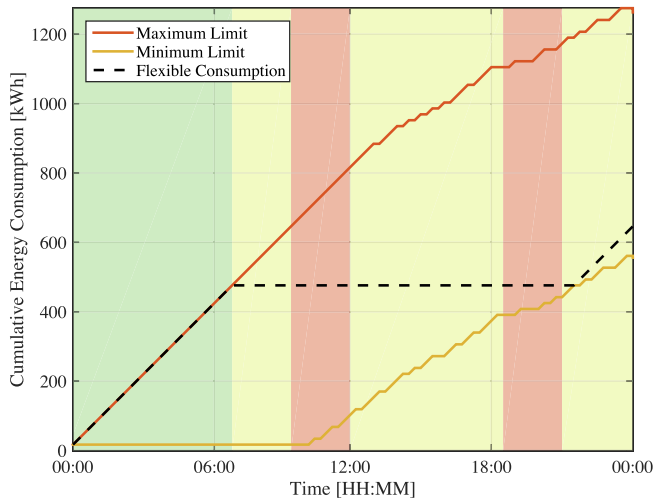
Despite the main objective of reducing operating costs related to energy consumption, the pumps ageing due to the number of pumping cycles is also a concern for the WSO, as reported in several meetings. Therefore, to tackle this issue, an additional cost is considered and added to (2) which is made dependent on the number pumping cycles during period T (in this work $T = 24$ h).

To apply the methodology described in Section 2, curves $E_{MAX}(n)$ and $E_{MIN}(n)$ are computed for period T using the water level model detailed in (1) and following a predictive approach considering the tank capacity of 6 000 m³, the maximum and minimum water levels of 4.9 and 1.2 m, respectively, the water consumption (see Fig. 6) and the pumping station demand (0 or 68 kW). A water inlet of 86 m³ is associated with every 15 min of continuous 68 kW power consumption. The flexible cumulative energy consumption $E_{FLEX}(n)$ is obtained also for period T by solving the problem described by (2)–(5) and using the previous costs as a penalty signal (see Fig. 8). For each time-step, the flexible cumulative energy consumption is given by:

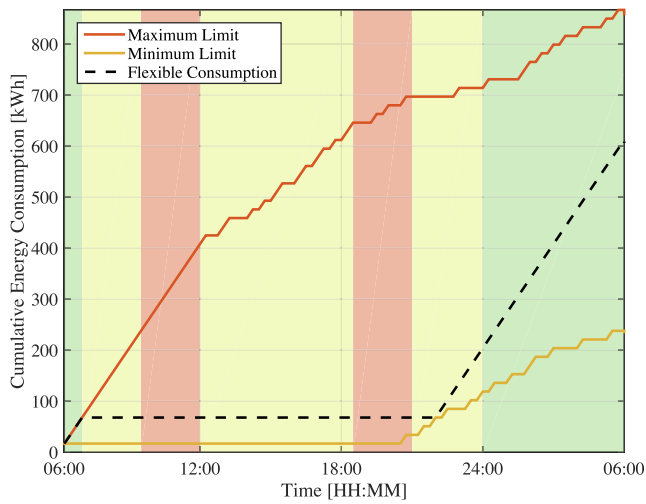
$$E_{FLEX}(n) = \frac{1}{4} \times \sum_{k=1}^n P(k) \quad (6)$$

where $P(k)$ refers to the pumping station power consumption from the beginning of the period of analysis, T , until time-step n . The $1/4$ factor in (6) is used to convert the 15-min resolution power consumption (kW) to energy (kWh). A genetic algorithm was used to solve the problem described by (2)–(5) using the WPSS power consumption P at each time-step, as the genes for each chromosome. To respect the real operation of the WPSS as imposed by the constraint expressed in (5), each gene can only assume the values 0 or 68 kW. Regarding the remaining constraints, the genetic algorithm only considers solutions that ensure $E_{FLEX}(n) \leq E_{max}(n)$ and $E_{FLEX}(n) \geq E_{min}(n)$ for $\forall n \in [0, T]$, as imposed by (3) and (4). Since a 15-min time-resolution was considered, each chromosome is therefore composed of 96 genes. Regarding crossover, mutation and elitism rates, these were tuned based on an extensive campaign of preliminary tests. The interested reader is referred to [25] for more details about the implementation of these algorithms.

To illustrate the application of this methodology, Fig. 9.a) shows



a)



b)

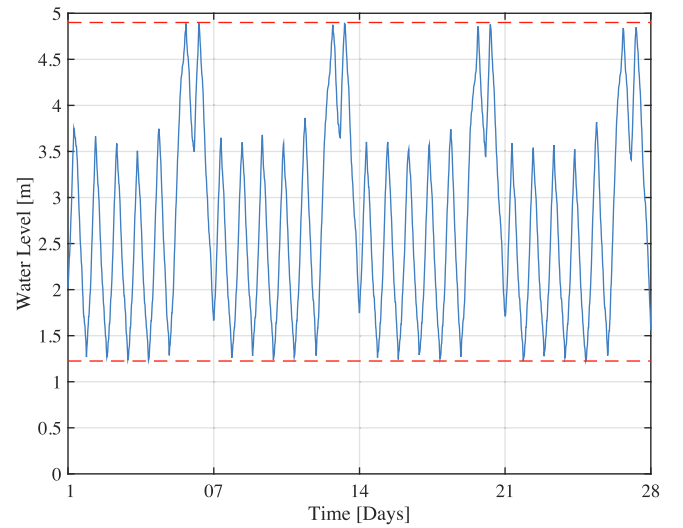
Fig. 9. Energy consumption limit curves and flexible energy consumption for February 28 when considering the internal use of energy flexibility. a) computed at 00:00 for the next 24 h. b) computed at 06:00 for the next 24 h.

$E_{MAX}(n)$, $E_{MIN}(n)$ and $E_{FLEX}(n)$ computed for the first time-step of the internal use of energy flexibility on February 28. Given the predicted water demand profile for the next 24 h, the pump operates until the beginning of the first half-peak period. Since the water level reached the minimum limit after the end of the second peak period it can be concluded that energy consumption during only off-peak periods would not be enough. Fig. 9.b) shows the same curves computed 6 h later. While $E_{FLEX}(n)$ presents a similar pattern (flat during the most expensive periods and with positive derivative in the remaining hours), $E_{MAX}(n)$ and $E_{MIN}(n)$ are considerably different. This results from different water levels at 00:00 and 06:00. In the first instant, the water level is lower as the pump has been operated for only 2.5 h since the end of the second peak period when the lower limit was reached. Therefore, $E_{MAX}(n)$ is constantly increasing for a longer period in Fig. 9 a) and $E_{MIN}(n)$ is flat for a shorter period. In both cases, the pump is requested to operate at $n = 0$.

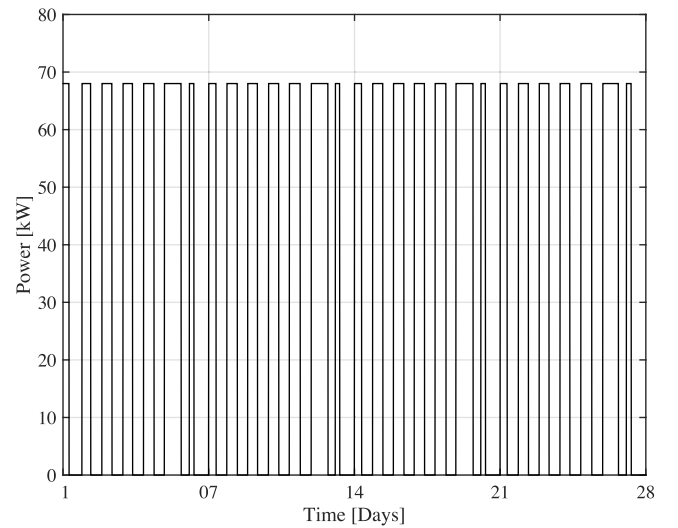
The operation of the WPSS follows a similar behaviour (when compared to Fig. 9) during the remaining business days of the

considered period, while during the weekend operation periods are higher due to the cheaper electricity prices, resulting in the water level and power consumption profiles shown in Fig. 10. It can be observed that water level limits are always respected and pumping cycles are significantly reduced when compared to the baseline operation. More precisely, according to the data reported in Fig. 5.b) these events are reduced by 57% (from 65 to 28). This is a direct consequence of affecting the cost function with the number of pumping cycles during period T , which forces the genetic algorithm to choose the chromosomes with fewer changes between genes among the population as best solutions.

The daily average profiles for the water level and power consumption during business days, which are obtained from Fig. 10, are shown in Fig. 11. These results show that the internal use of energy flexibility allows the WPSS to completely avoid operation during peak periods and to transfer most of its consumption to off-peak periods. Besides the drastic reduction of pumping cycles, the transfer of operation to the cheapest periods results in 16.4% cost reduction during the period of analysis when compared to the baseline scenario (from € 2000 to €

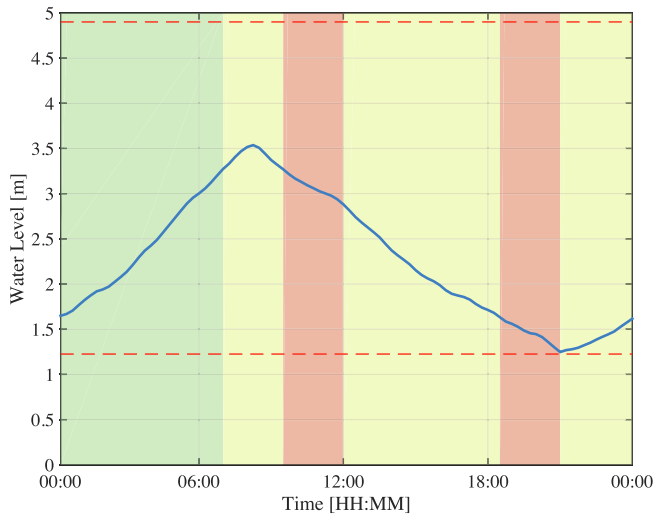


a)

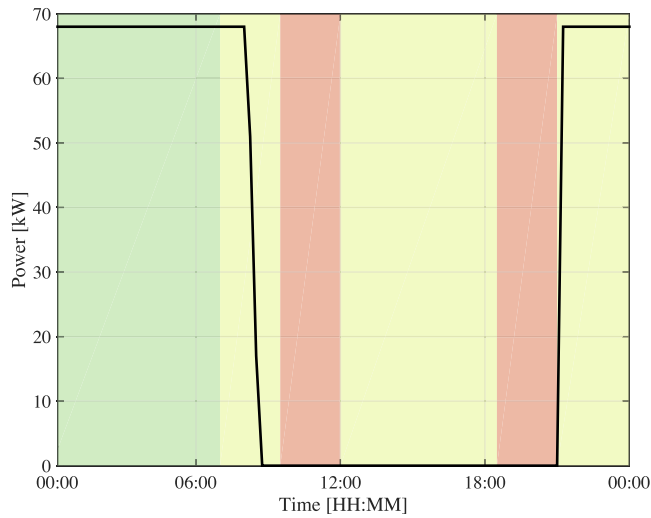


b)

Fig. 10. Modified operation, due to the internal use of energy flexibility, from 28/02/2017 to 27/03/2017. a) tank water level. b) electrical energy consumption.



a)



b)

Fig. 11. Daily average water level (a) and power consumption (b) profiles associated with the internal use of energy flexibility during weekdays.

1672). Fig. 12 presents the costs associated with each period for both the baseline and modified operation due to the internal use of energy flexibility. Apart from the costs reported in this figure, both scenarios are subjected to the same cost of contracted power, which is defined based on the maximum 15-min resolution peak load of the last 12 months. Value-added tax and other taxes are also not represented in Fig. 12. It is important to note that energy consumption during half-peak can be completely avoided if both pumps operate at night. However, such operation would be characterised by high hydraulic losses and increased equipment ageing and for that reason the WSO specifies that both pumps should only operate simultaneously during emergency events.

4.2. External use of energy flexibility to support power system operation

In this study, the external use of energy flexibility to support power system operation is illustrated in the context of the first two wind generation curtailment events registered in Portugal, which occurred on March 12 and 13, 2017. In the first one, on March 12, the Portuguese Transmission System Operator (TSO) curtailed 90 MW of wind generation from 15:00 to 18:00. Regarding the second event, it started at 21:25

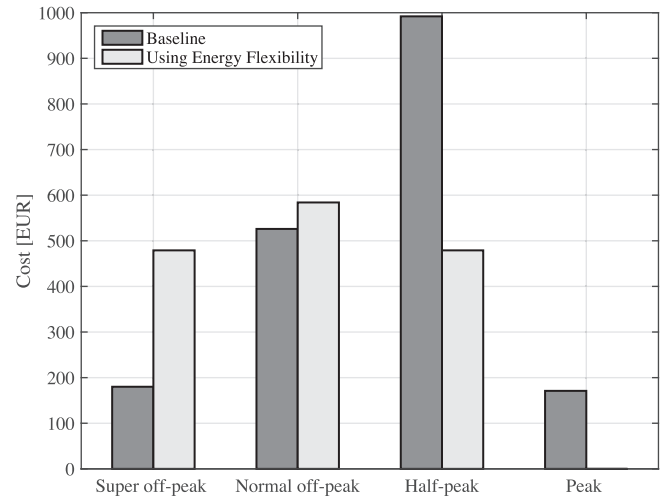


Fig. 12. Distribution of electrical energy costs considering the baseline operation and the modified operation using energy flexibility.

on March 12 and finished at 04:26 on March 13, where wind generation was curtailed in two phases. In the first one, from 21:25 to 22:02, wind generation was decreased by 80 MW while in the second phase, from 22:02 to 04:26, an additional 80 MW curtailment was applied. Fig. 13 presents the Portuguese power system production mix during these two days, together with the Portuguese electrical energy demand, where the red dashed lines are associated with the start and end instants of each curtailment period. Considering the information reported by the Portuguese TSO ([26]), wind-based generation was supplying more than 90% of the Portuguese electrical energy demand at the beginning of these curtailment periods (wind and other renewable generation were exceeding the electrical energy demand by 950 MW). Additionally, some factors contributed to the need of curtailment, namely: i) electrical energy demand was relatively low, as it is typical for a Sunday with moderate temperatures; ii) no thermal power plants were contributing to the production mix; iii) there was no electrical energy import from Spain; iv) hydro generation was already at a minimum level; and v) all pumped-storage hydropower was already supporting the operation of the power system.

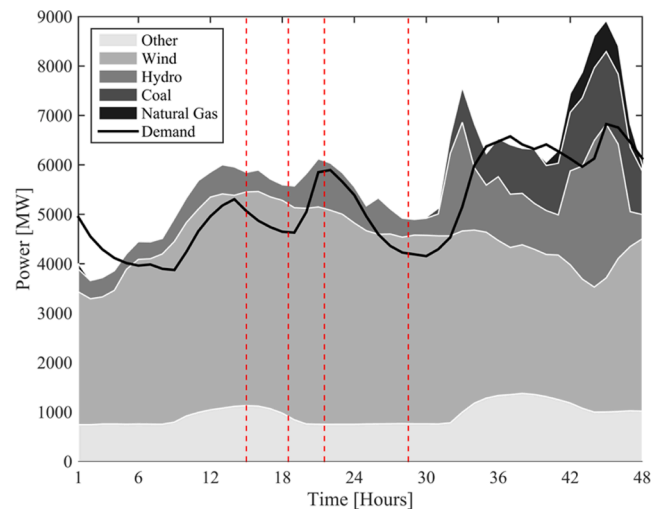


Fig. 13. Electrical energy production mix and consumption from 12/03/2017 to 13/03/2017.

In this case, the energy flexibility provided by the WPSS is used to force electrical energy consumption during the curtailment events, considering a penalty signal with zero or positive values outside the curtailment period and negative values during these events (see (2)). To apply this strategy, the external entity using the referred flexibility receives updated $E_{MAX}(n)$ and $E_{MIN}(n)$ curves at each time-step and communicates $E_{FLEX}(0)$ to the WPSS. The optimization problem described by (2)–(5) is solved using the genetic algorithm referred in Section 4.1.

Fig. 14 presents the energy demand profile of the WPSS for both internal and external use of flexibility, where the start and end instants of the curtailment periods are identified by the red dashed lines. Since the first curtailment event occurred on a Sunday, the water level was relatively high and therefore WPSS operation was stopped before the end of the event, as it can be observed in Fig. 15. When the second curtailment event starts, the water level is still high and therefore WPSS operation is maintained until the reservoir is full again. After the end of the second curtailment event, it is assumed that the WPSS returns to its normal operation, where energy flexibility is internally used to reduce energy costs. The fact that these curtailment events occurred on a Sunday deeply influences the flexibility provided by the WPSS, since during its normal operation water levels are already relatively high. In case these events have been registered during business days the contribution of the WPSS to increase energy demand would be considerably higher.

It is important to note that the external use of the available energy flexibility may impact the WPSS energy costs due to the respective modifications in the power consumption profile when compared to the internal use, as depicted in Fig. 14. Therefore, although the development of specific business models is out of the scope of this paper, these should be considered when using the proposed methodology to compensate possible losses related to higher energy costs that are supported by the WSO. For instance, according to the Portuguese legislation (Decree-Law Nr. 172/2006) entities responsible for large loads can be enrolled in DR programs, where the TSO can interrupt the respective power consumption profiles during certain periods of time to support the power system operation. In this case, the entities responsible for the large loads receive a fixed monetary compensation to be enrolled in the DR programs plus a variable compensation that depends on the number and duration of load interruptions.

There are several benefits reported for the use of DR schemes. As an example, they can be divided according to the type of those benefits, such as participant, market-wide, reliability and market performance [27]. In this study, the existing energy flexibility is used by the TSO to support the operation of the power system during the first wind curtailment event registered in Portugal. Nevertheless, other external

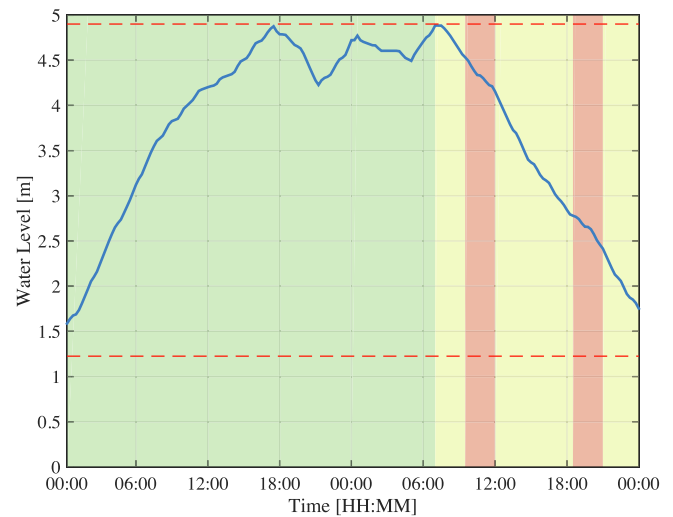


Fig. 15. Water level from 12/03/2017 to 13/03/2017.

entities can also be considered. For instance, aggregators could use the energy flexibility provided by WPSS to participate in existing power exchange markets and therefore extract benefits from such participation, which would be shared with the WSO so its participation could be encouraged (e.g. economic benefits). In this case, it should be noted that several business models have been proposed for aggregators of energy flexibility [28], with focus on, e.g., electric vehicles [29] or buildings [30]. Alternatively, the flexibility can be used by grid operators directly [31] or indirectly through grid-based penalty signals [32], allowing system integration to leverage energy flexibility across different energy sectors [33]. Another example refers to the use of the available energy flexibility in renewable energy communities to achieve specific local objectives that bring benefits for both the community and the DSO (e.g., increase renewable energy self-consumption in order to reduce possible negative impacts due to excessive reverse power flow, such as increased ageing of distribution transforms [34]). This is of particular importance when considering recent legislation at European level targeting the adoption of renewable energy communities (see, for instance, Decree-Law Nr. 162/2019 for Portugal).

5. Conclusion and future work

The integration of renewable energy resources into existing power systems or the increasing electrification of energy demand calls for the development of innovative solutions to tackle possible negative impacts. In this context, the energy flexibility provided by water pumping and storage systems can be used by external entities to support the operation of power systems. At the same time, WPSS operators are also interested in using the referred flexibility internally to achieve certain objectives, such as reducing energy consumption costs.

This paper presents a methodology to support the use of the energy flexibility provided by WPSS, both at internal and external levels, and the respective application to a case study. This methodology uses cumulative energy consumption curves to represent the maximum and minimum energy limits and the associated flexible energy consumption, which is defined in order to reduce the penalty imposed on the WPSS (e.g. electrical energy price). By following this approach, the external entity using the available energy flexibility does not require specific knowledge about the physical parameters of the WPSS to define the desired energy demand profile but only the referred limit curves and the possible power consumption values.

For the WPSS considered in this work, the collected results show that the application of the referred methodology reduces electrical energy

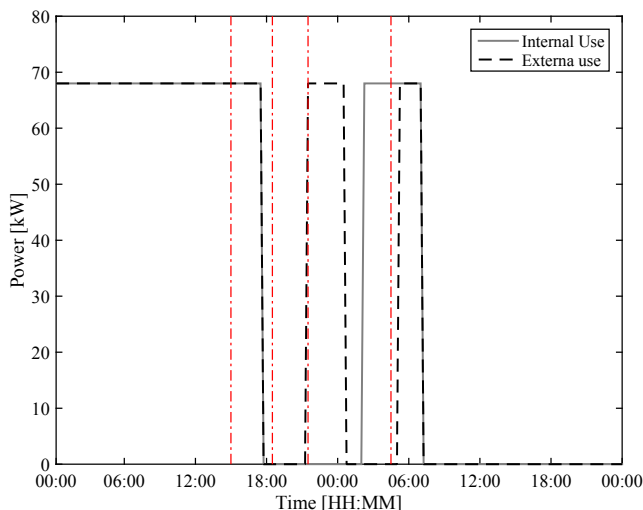


Fig. 14. Power consumption from 12/03/2017 to 13/03/2017.

costs by 16.4% compared to the baseline operation. Additionally, this internal use of energy flexibility reduces pumping cycles by 57%, which contributes to reduce equipment ageing. The external use of energy flexibility is illustrated in the context of the first wind generation curtailment registered in Portugal. While the control of this single WPSS would not avoid the referred curtailment due to the involved amount of power, the results show that the WPSS operation could be modified to increase energy consumption during this period of excessive wind generation and therefore support the power system operator. The aggregation of several WPSS and the use of the respective aggregated energy flexibility by external entities constitute an important research work left open by this study. The development of business models to support the relationship between the internal and external use of energy flexibility should also be taken into consideration.

CRedit authorship contribution statement

Rui Amaral Lopes: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Rune Grønberg Junker:** Investigation, Writing - original draft, Writing - review & editing. **João Martins:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing. **João Murta-Pina:** Investigation, Writing - original draft. **Glenn Reynders:** Writing - original draft, Writing - review & editing. **Henrik Madsen:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Electricity costs during each time-of-use period were collected from the electricity bill provided by the WSO for the period of analysis (from 28/02/2017 to 27/03/2017) and result from negotiations with electric energy providers as presented in Table A1. The referred time-of-use periods are set by Portuguese legislation [24] (see Table A1).

Table A1

Time-of-use periods and tariffs (before taxes) associated with weekly-cycles contracts.

Day	Period	Weekly-cycle	Cost [€/kWh]
Business days	Peak	09:30 – 12:00	0.1063
		18:30 – 21:00	
		07:00 – 09:30	
	Half-peak	12:00 – 18:30	0.0974
		21:00 – 00:00	
		00:00 – 02:00	
Saturday	Normal off-peak	06:00 – 07:00	0.0648
		02:00 – 06:00	
		02:00 – 06:00	
	Super off-peak	02:00 – 06:00	0.0630
		–	
		–	
Sunday	Peak	09:30 – 13:00	0.0974
		18:30 – 22:00	
		00:00 – 02:00	
	Normal off-peak	06:00 – 09:30	0.0648
		13:00 – 18:30	
		22:00 – 00:00	
Sunday	Super off-peak	02:00 – 06:00	0.0630
		–	
		–	
	Peak	–	–
		–	
		–	
Sunday	Half-peak	00:00 – 02:00	0.0648
		06:00 – 00:00	
		02:00 – 06:00	
	Super off-peak	02:00 – 06:00	0.0630
		–	
		–	

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